

# EXPERIMENT NO 14

**Determination and plotting of Polarization in different Dielectric Medium with their relative permittivity and to find electric susceptibility.**

**Aim:**

To determine and plot polarization in different dielectric medium with their relative permittivity and to find electric susceptibility.

**Software required:**

- **SCILAB version 6.0.1 Program:**

```
EXP 14.sci (C:\Users\user\OneDrive\Documents\EXP 14.sci) - SciNotes
File Edit Format Options Window Execute ?
EXP 14.sci (C:\Users\user\OneDrive\Documents\EXP 14.sci) - SciNotes
EXP 14.sci
5 epsilon0 = 8.854e-12;
6
7 // Input values
8 ex = input("Enter the relative permittivity (ex): ");
9 field = input("Enter the Electric Field (V/m) : ");
10
11 // Electric susceptibility
12 chi = ex - 1;
13
14 // Polarisation
15 polarisation = chi * epsilon0 * field;
16
17 // Display results
18 sprintf("Electric Susceptibility (chi) = %.6f\n", chi);
19 sprintf("Polarisation (P) = %.6e C/m^2\n", polarisation);
20
21
22 // Values for plotting
23 n = linspace(0, 10, 50);
24
25 // Susceptibility is constant for a fixed ex
26 y = chi * ones(1, length(n));
27
28 // Polarisation as electric field varies
29 E_values = 0.1 * n;
30 z = chi * epsilon0 * E_values;
31
32 // Create figure
33 figure(1);
34 clf();
35
36 // Graph 1: Electric Susceptibility
37 subplot(2, 1, 1);
38 plot(n, y);
39 plot(n, z);
40 xlabel("Electric Field (V/m)");
41 ylabel("Electric Susceptibility");
42 title("ELECTRIC SUSCEPTIBILITY");
43 grid();
44
45 // Graph 2: Polarisation
46 subplot(2, 1, 2);
47 plot(n, E_values, z);
48 xlabel("Electric Field (V/m)");
49 ylabel("Polarisation (C/m^2)");
50 title("POLARIZATION");
51 grid();
52
```

**Output:**

Scilab 2026.1.0 Console

File Edit Control Applications ?

File Browser

C:\Users\user\Documents\

Name

Documents

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ex 6.sd

EXP 14.sd~

Enter the relative permittivity (er):  $6 \times 10^{-7}$

Enter the Electric Field (V/m):  $4 \times 10^{-7}$

Electric Susceptibility ( $\chi_1$ ) = 52.000000

Polarization (P) =  $1.519346e-08$  C/m<sup>2</sup>

exec: Wrong number of output argument(s): 0 expected.

-->

Variable Browser

| Name         | Value    | Type   | Visibility | Memory |
|--------------|----------|--------|------------|--------|
| epsilon0     | 8.85e-12 | Double | local      | 216 B  |
| epsilon_air  | 8.85e-12 | Double | local      | 216 B  |
| epsilon_v... | 8.85e-12 | Double | local      | 216 B  |
| epsilon_...  | 7.08e-10 | Double | local      | 216 B  |
| eta_air      | 377      | Double | local      | 216 B  |
| eta_vacu...  | 377      | Double | local      | 216 B  |
| eta_water    | 42.1     | Double | local      | 216 B  |
| mu0          | 1.26e-06 | Double | local      | 216 B  |
| mu_air       | 1.26e-06 | Double | local      | 216 B  |

Command History

-- 10/08/2026 14:33:17 -- //

-- 10/08/2026 14:50:34 -- //

-- 24/08/2026 04:27:36 -- //

-- 24/08/2026 04:29:27 -- //

3\*10-7

6

2

5

3

4

5

3

4

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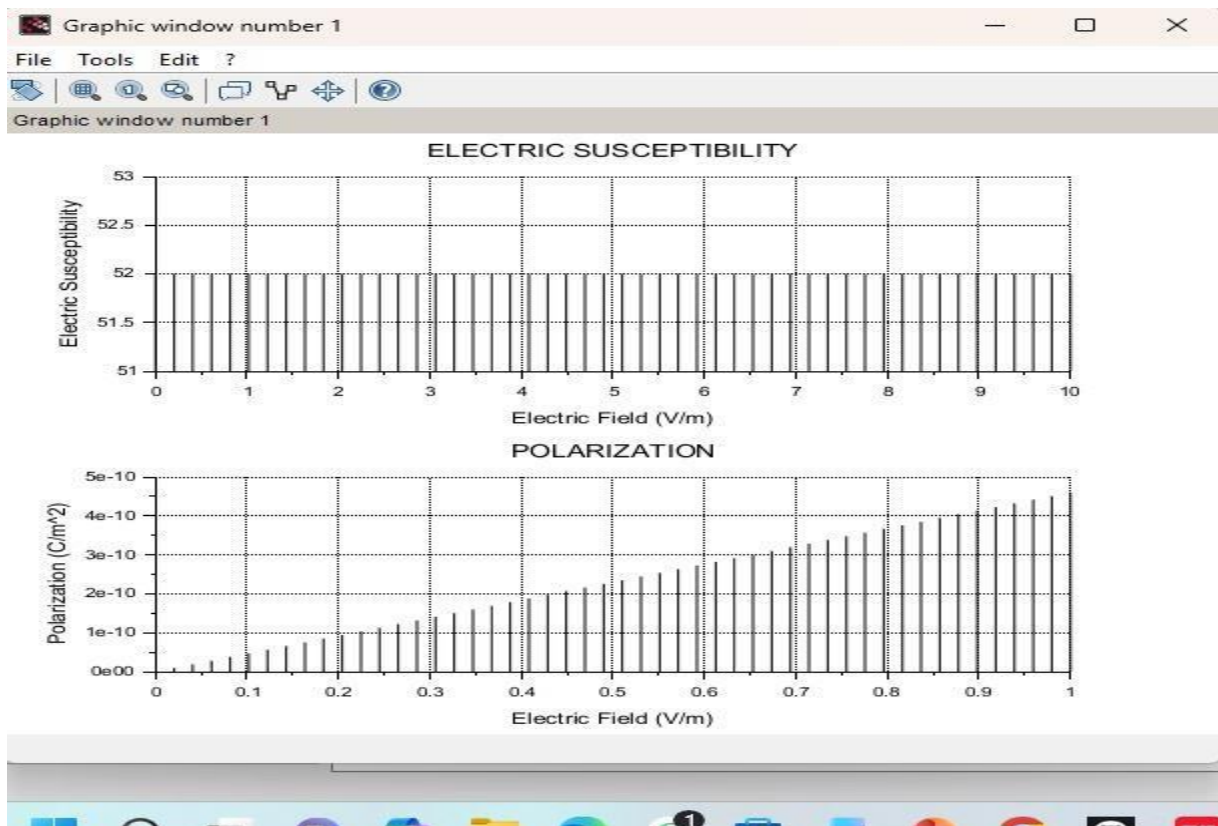
Giving back support or use cases is appreciated.

- Add and improve functions on [GiteLab](#).
- View and edit [help pages](#).
- Publish toolboxes on [ATOMS](#).
- Post questions on [Discourse](#).

ENG IN

5:38 AM

8/24/2026



Theoretical calculations:

Ex. No. 14

$$\epsilon_r = 5 \times 10^{-6}$$

$$\text{electric field} = 3 \times 10^{-2}$$

1. Electric susceptibility ( $\chi$ ):

$$\chi = 5 \times 10^{-6} - 1 = -0.999995$$

2. Electric polarization ( $P$ ):

$$P = (-0.999995) \times (8.854 \times 10^{-12}) \times (3 \times 10^{-2})$$

$$P = -0.999995 \times (2.6562 \times 10^{-13}) = -2.6562 \times 10^{-13}$$

**Case 1: To determine and plot the polarization and electric susceptibility of electromagnetic waves in three dielectric media: Air, Water, and Glass, using their relative permittivity.**

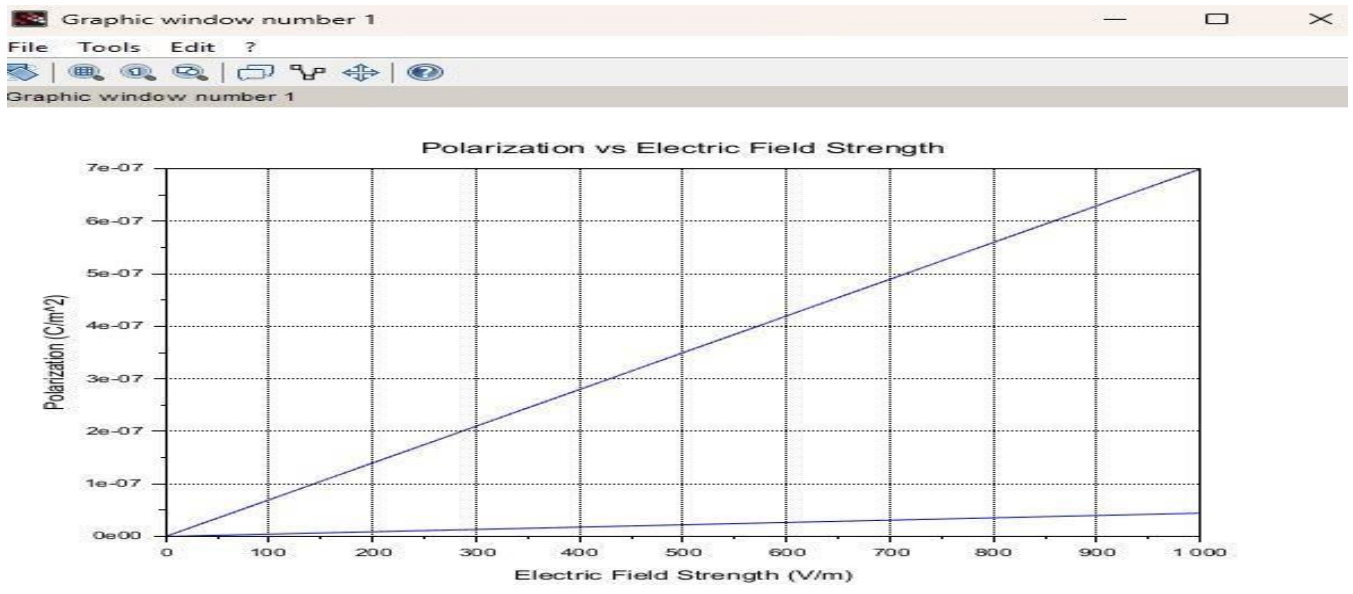
**Program:**

```

EXP 14.sci (C:\Users\user\OneDrive\Documents\EXP 14.sci) - SciNotes
File Edit Format Options Window Execute ?
EXP 14.sci (C:\Users\user\OneDrive\Documents\EXP 14.sci) - SciNotes
EXP 14.sci x
1 clear
2 clear;
3
4 // Permittivity of free space
5 epsilon_0 = 8.854e-12;
6
7 // Electric field strength
8 E = linspace(0, 1000, 100);
9
10 // Relative permittivity
11 er_air = 1;
12 er_water = 80;
13 er_glass = 4;
14
15 // Electric susceptibility
16 chi_air = er_air - 1;
17 chi_water = er_water - 1;
18 chi_glass = er_glass - 1;
19
20 // Polarisation
21 P_air = epsilon_0 * chi_air * E;
22 P_water = epsilon_0 * chi_water * E;
23 P_glass = epsilon_0 * chi_glass * E;
24
25 // Create graph
26 h = figure;
27 title('');
28
29 // Plot Air
30 plot(E, P_air);
31
32 // Plot Water
33 plot(E, P_water);
34
35 // Plot Glass
36 plot(E, P_glass);
37
38 // Labels
39 label('Electric Field Strength (V/m)');
40 label('Polarisation (C/m^2)');
41 title('Polarisation vs Electric Field Strength');
42
43 // Grid
44 grid();
45

```

**Output:**



**Case 2:** To calculate and plot the polarization and electric susceptibility in Silicon, Teflon, and Paper, and understand how their relative permittivity affects the polarization.

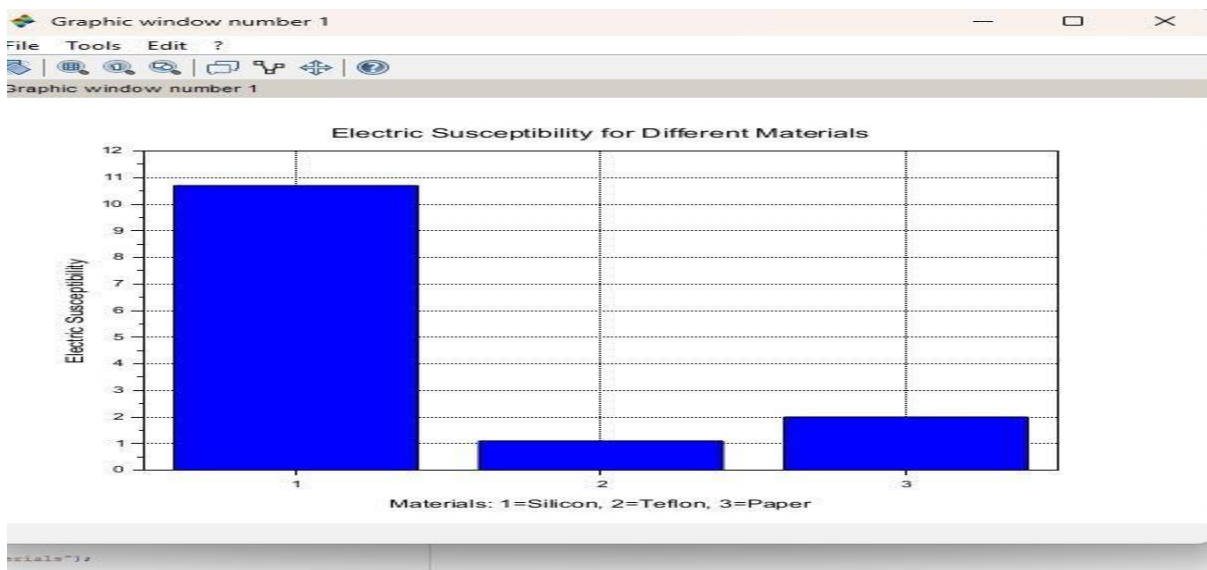
**Program:**

```

EXP 14.sci (C:\Users\user\OneDrive\Documents\EXP 14.sci) - SciNotes
File Edit Format Options Window Execute ?
EXP 14.sci (C:\Users\user\OneDrive\Documents\EXP 14.sci) - SciNotes
EXP 14.sci
1 clear;
2 clear;
3
4 // Constant
5 epsilon0 = 8.854e-12;
6
7 // Relative permittivity
8 epsilon_r = [11.7 2.1 3.0];
9
10 // Electric susceptibility
11 chi_e = epsilon_r - 1;
12
13 // Electric field
14 E = 1e6;
15
16 // Polarisation
17 polarisation = epsilon0 .* chi_e .* E;
18
19 // Display values
20
21 disp('Electric Susceptibility:');
22 disp(chi_e);
23
24 disp('Polarisation (C/m^2):');
25 disp(polarisation);
26
27 // Graph 1 -- Electric Susceptibility
28
29 clf(1);
30
31 bar(chi_e);
32
33 xlabel('Materials: 1=Silicon, 2=Teflon, 3=Paper');
34 ylabel('Electric Susceptibility');
35 title('Electric Susceptibility for Different Materials');
36 legend();
37
38 // Graph 2 -- Polarisation
39
40 clf(2);
41
42 bar(polarisation);
43
44 xlabel('Materials: 1=Silicon, 2=Teflon, 3=Paper');
45 ylabel('Polarisation (C/m^2)');
46 title('Polarisation for Different Materials');
47 legend();
48

```

## Output:



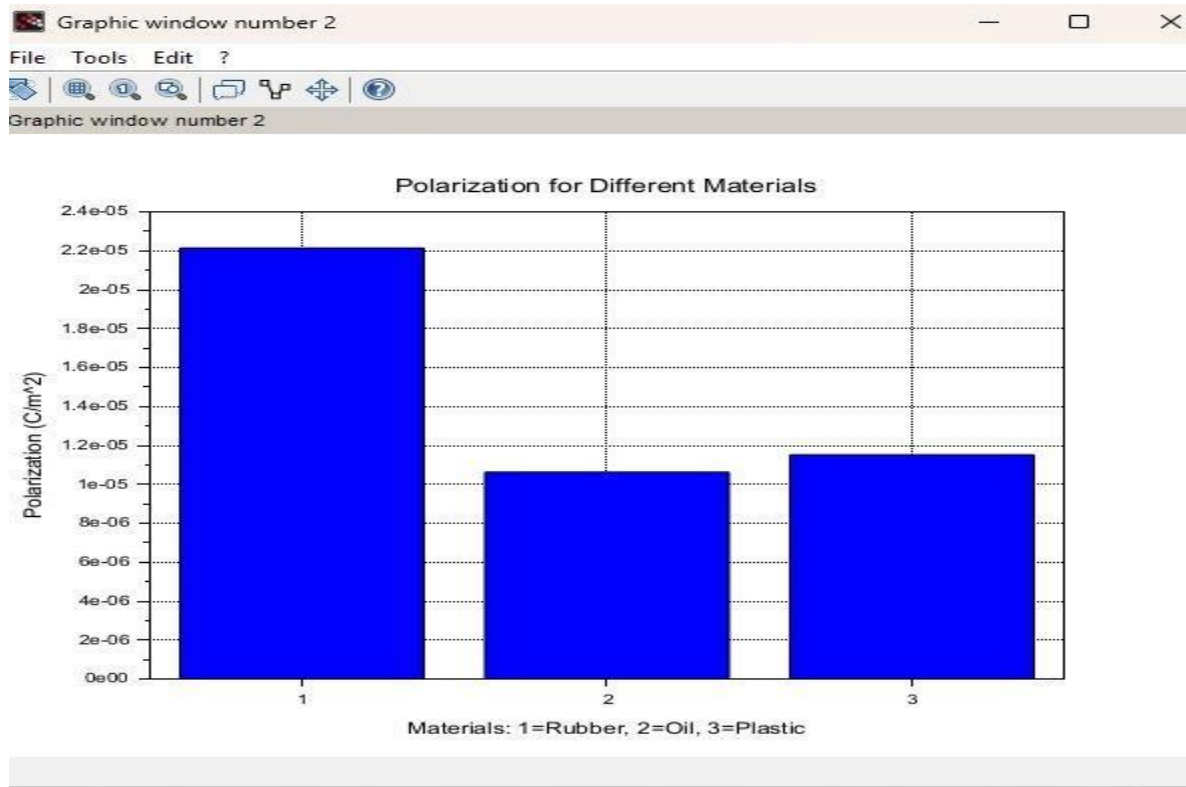
**Case 3:** To explore how different dielectric materials (Rubber, Oil, and Plastic) affect the polarization and electric susceptibility, and to compare their relative permittivity.

## Program:

```
EXP 14.sci (C:\Users\user\OneDrive\Documents\EXP 14.sci) - SciNotes
File Edit Format Options Window Execute ?
EXP 14.sci (C:\Users\user\OneDrive\Documents\EXP 14.sci) - SciNotes
EXP 14.sci X

1 clc;
2 clear;
3
4 // Constant
5 epsilon0 = 8.854e-12;
6
7 // Relative permittivity
8 epsilon_r = [2.5 2.2 2.2];
9
10 // Electric susceptibility
11 chi_e = epsilon_r - 1;
12
13 // Applied electric field
14 E = 1e6;
15
16 // Polarisation
17 polarisation = epsilon0 .* chi_e .* E;
18
19
20 // =====
21 // DISPLAY RESULTS
22 // =====
23
24 disp("Electric Susceptibility:");
25 disp(chi_e);
26
27 disp("Polarisation (C/m^2):");
28 disp(polarisation);
29
30
31 // =====
32 // GRAPH 1: ELECTRIC SUSCEPTIBILITY
33 // =====
34
35 scf(1);
36 clf();
37
38 bar(chi_e);
39
40 xlabel("Materials: 1=Rubber, 2=Oil, 3=Plastic");
41 ylabel("Electric Susceptibility");
42 title("Electric Susceptibility for Different Materials");
43
44
45
46 // =====
47 // GRAPH 2: POLARISATION
48 // =====
49
50
51 scf(2);
52 clf();
53
54 bar(polarisation);
55
56 xlabel("Materials: 1=Rubber, 2=Oil, 3=Plastic");
57 ylabel("Polarisation (C/m^2)");
58 title("Polarisation for Different Materials");
59
60 xgrid();
61
```

Output:



**Case 1:** Examine the effect of length of a ferromagnetic material on the magnetic flux density ( $B$ ) for the following conditions

- ☐ The material is ferromagnetic with relative permeability ( $\mu_r$ ) of 500.
- ☐ The magnetic field intensity ( $H$ ) is held constant at 100 A/m.
- ☐ The length of the material varies from 0.1 m to 2 m.



## Program:

```
EXP 14 sci (C:\Users\user\OneDrive\Documents\EXP 14 sci) - SciNotes
File Edit Format Options Window Execute ?
EXP 14 sci (C:\Users\user\OneDrive\Documents\EXP 14 sci) - SciNotes
EXP 14.sci
1 clear;
2 clear;
3 %;
4
5 // Constants
6 mu_r = 500; // Relative permeability
7 H = 100; // Magnetic field intensity (A/m)
8 mu_0 = 4 * pi * 10^-7; // Permeability of free space (H/m)
9
10 // Lengths from 0.1 m to 2 m
11 lengths = 0.1:0.1:2;
12
13 // Initialize array for magnetic flux density
14 B_values = zeros(1, length(lengths));
15
16 // Calculate magnetic flux density
17 for i = 1:length(lengths)
18     B_values(i) = mu_r * mu_0 * H;
19 end
20
21 // Display results
22 disp('Length (m) ----- Magnetic Flux Density (T)');
23
24 for i = 1:length(lengths)
25     sprintf('%1.1f-----%.6f\n', lengths(i), B_values(i));
26 end
27
28 // Plot the results
29 plot(lengths, B_values, 'o');
30 xlabel('Length of Ferromagnetic Material (m)');
31 ylabel('Magnetic Flux Density B (T)');
32 title('Effect of Length on Magnetic Flux Density');
33 grid();
34
```

## Output:

Scilab 2026.1.0 Console

File Edit Control Applications ?

File Browser

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Documents

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- My Videos
- ex 6.sci
- EXP 14.sci

File/directory filter

Scilab 2026.1.0 Console

```
"Length (m)      Magnetic Flux Density (T)"
0.1      0.062832
0.2      0.062832
0.3      0.062832
0.4      0.062832
0.5      0.062832
0.6      0.062832
0.7      0.062832
0.8      0.062832
0.9      0.062832
1.0      0.062832
1.1      0.062832
1.2      0.062832
1.3      0.062832
1.4      0.062832
1.5      0.062832
1.6      0.062832
1.7      0.062832
1.8      0.062832
1.9      0.062832
2.0      0.062832
-->
```

Variable Browser

| Name     | Value     | Type   | Visibility | Memory |
|----------|-----------|--------|------------|--------|
| B_values | 1x20      | Double | local      | 368 B  |
| H        | 100       | Double | local      | 216 B  |
| i        | 20        | Double | local      | 216 B  |
| lengths  | 1x20      | Double | local      | 368 B  |
| mu_0     | 1.256e-06 | Double | local      | 216 B  |
| mu_r     | 500       | Double | local      | 216 B  |

Command History

```
4
2
-2
5
3
4
5
```

EXP 14.sci (C:\Users\user\OneDrive\Documents\EXP 14.sci) - SciNotes

File Edit Format Options Window Execute ?

EXP 14.sci (C:\Users\user\OneDrive\Documents\EXP 14.sci) - SciNotes

EXP 14.sci

```
1 clear;
2 clear;
3 %;
4
5 // Constants
6 mu_r = 500; // Relative permeability
7 H = 100; // Magnetic field intensity (A/m)
8 mu_0 = 4 * pi * 10^-7; // Permeability of free space (H/m)
9
10 // Lengths from 0.1 m to 2 m
11 lengths = 0.1:0.1:2;
12
13 // Initialize array for magnetic flux density
14 B_values = zeros(1, length(lengths));
15
16 // Calculate magnetic flux density
17 for i = 1:length(lengths)
18     B_values(i) = mu_r * mu_0 * H;
19 end
20
21 // Display results
22 disp('Length (m) ----- Magnetic Flux Density (T)');
23
24 for i = 1:length(lengths)
25     sprintf('%1.1f-----%.6f\n', lengths(i), B_values(i));
26 end
27
28 // Plot the results
29 plot(lengths, B_values, 'o');
30 xlabel('Length of Ferromagnetic Material (m)');
31 ylabel('Magnetic Flux Density B (T)');
32 title('Effect of Length on Magnetic Flux Density');
33 grid();
34
```

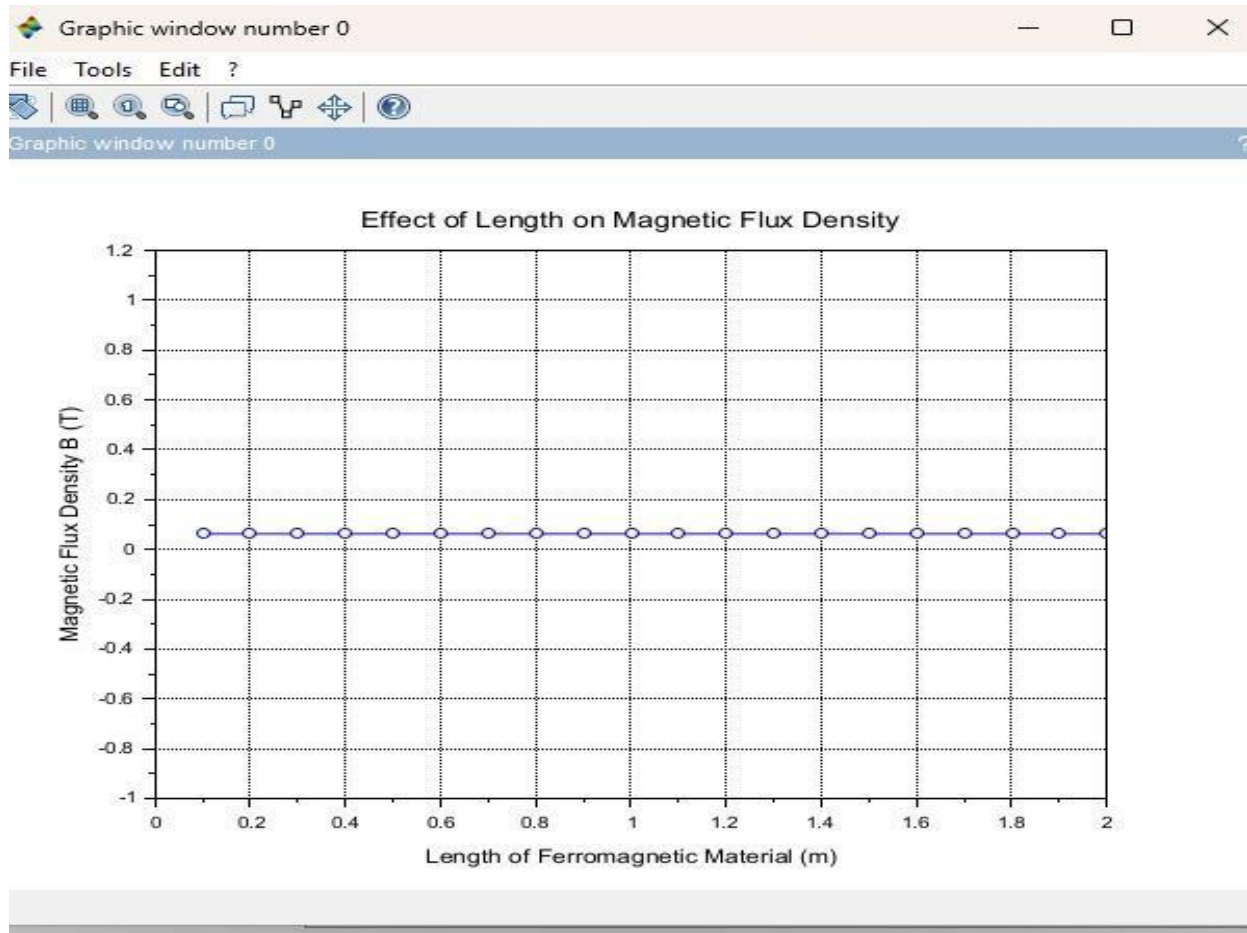
Snipping Tool

Screenshot copied to clipboard  
Automatically saved to screenshots folder.

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11:29 AM  
8/30/2026





**Case 2:** To investigate how the magnetic flux density ( $B$ ) in a ferromagnetic material changes with different widths at a constant length and magnetic field intensity.

Conditions:

- Material: Soft iron (same as in Case Study 1)
- Length ( $L$ ): Constant at 10 cm
- Magnetic Field Intensity ( $H$ ): Constant at 1000 A/m
- Width ( $W$ ): Varies from 2 cm to 10 cm (i.e., 2 cm, 4 cm, 6 cm, 8 cm, 10 cm)

## Program:

```
EXP 14.sci (C:\Users\user\OneDrive\Documents\EXP 14.sci) - SciNotes
File Edit Format Options Window Execute ?
EXP 14.sci (C:\Users\user\OneDrive\Documents\EXP 14.sci) - SciNotes
EXP 14.sci x
1 clear;
2 clear;
3 %%%
4
5 // Constants
6 L = 0.1; // Length in meters (10 cm)
7 H = 1000; // Magnetic field intensity (A/m)
8 mu_0 = 4 * pi * 10^(-7); // Permeability of free space (H/m)
9
10 // Widths in meters (2 cm to 10 cm)
11 widths = [0.02, 0.04, 0.06, 0.08, 0.10];
12 num_widths = length(widths);
13
14 // Relative permeability of soft iron
15 mu_r = 1000;
16
17 // Initialise array
18 E_values = zeros(1, num_widths);
19
20 // Calculate magnetic flux density
21 for i = 1:num_widths
22     E_values(i) = mu_0 * mu_r * H;
23 end
24
25 // Display results
26 disp("Width (m) ----- Flux Density (T)");
27
28 for i = 1:num_widths
29     fprintf("%5.2f-----%5.6f\n", widths(i), E_values(i));
30 end
31
32 // Plot the results
33 plot(widths, E_values, '-o');
34
35 xlabel("Width of Ferromagnetic Material (m)");
36 ylabel("Magnetic Flux Density B (T)");
37 title("Effect of Width on Magnetic Flux Density");
38
39 xgrid();
40
```

## Output:

Scilab 2026.1.0 Console

File Edit Control Applications ?

File Browser

C:\Users\user\Documents\

Name

- Documents
- My Music
- My Pictures
- My Videos
- ex 6.sd
- EXP 14.sci~

File/directory filter

Scilab 2026.1.0 Console

```
"Width (m)      Flux Density (T)"
0.02            1.256637
0.04            1.256637
0.06            1.256637
0.08            1.256637
0.10            1.256637
--> |
```

Variable Browser

| Name       | Value    | Type   | Visibility | Memory |
|------------|----------|--------|------------|--------|
| E_values   | 1x5      | Double | local      | 248 B  |
| H          | 1e+03    | Double | local      | 216 B  |
| L          | 0.1      | Double | local      | 216 B  |
| i          | 5        | Double | local      | 216 B  |
| mu_0       | 1.26e-06 | Double | local      | 216 B  |
| mu_r       | 1e+03    | Double | local      | 216 B  |
| num_widths | 5        | Double | local      | 216 B  |
| widths     | 1x5      | Double | local      | 248 B  |

Command History

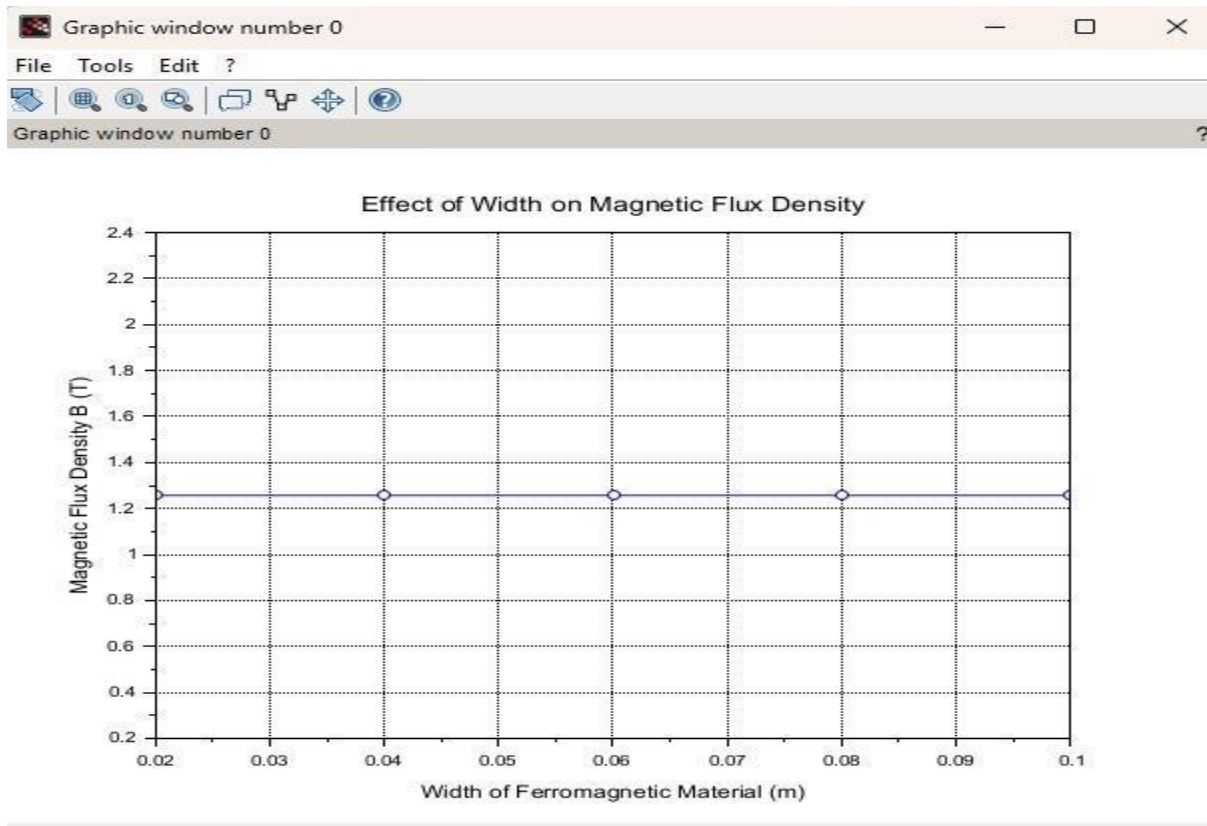
```
1 clear;
2 clear;
3 %%%
4
5 // Constants
6 L = 0.1; // Length in meters (10 cm)
7 H = 1000; // Magnetic field intensity (A/m)
8 mu_0 = 4 * pi * 10^(-7); // Permeability of free space (H/m)
9
```

Snipping Tool

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8/30/2026



**Case 3:** To examine how the magnetic flux density ( $B$ ) in a ferromagnetic material changes with different values of magnetic field intensity at constant length and width.

Conditions:

- Material: Soft iron (same as in the previous studies)
- Length ( $L$ ): Constant at 10 cm
- Width ( $W$ ): Constant at 5 cm
- Magnetic Field Intensity ( $H$ ): Varies from 200 A/m to 2000 A/m (i.e., 200 A/m, 500 A/m, 1000 A/m, 1500 A/m, 2000 A/m)

## Program:

```
EXP 14.sci (C:\Users\user\OneDrive\Documents\EXP 14.sci) - SciNotes
File Edit Format Options Window Execute ?

EXP 14.sci (C:\Users\user\OneDrive\Documents\EXP 14.sci) - SciNotes
EXP 14.sci X

1  clc;
2  clear;
3  clf();
4
5  // Define constants
6  L = 0.1; // Length in meters (10 cm)
7  W = 0.05; // Width in meters (5 cm)
8
9  H_values = [200, 500, 1000, 1500, 2000]; // H in A/m
10
11 mu_0 = 4 * pi * 10^(-7); // Permeability of free space (H/m)
12 mu_r = 1000; // Relative permeability of soft iron
13
14 // Initialise array for magnetic flux density
15 B_values = zeros(1, length(H_values));
16
17 // Calculate B for each H value
18 for i = 1:length(H_values)
19     H = H_values(i);
20     mu = mu_0 * mu_r; // Absolute permeability
21     B_values(i) = mu * H; // B = mu * H
22 end
23
24 // Display results
25 disp('Magnetic Field Intensity (A/m) --- Magnetic Flux Density (T)');
26
27 for i = 1:length(H_values)
28     fprintf('%10.0f --- %10.0f\n', H_values(i), B_values(i));
29 end
30
31 // Plot the results
32 plot(H_values, B_values, 'o');
33
34 xlabel('Magnetic Field Intensity H (A/m)');
35 ylabel('Magnetic Flux Density B (T)');
36 title('Magnetic Flux Density vs Magnetic Field Intensity for Soft Iron');
37
38 grid();
39
```

## Output:

Scilab 2026.1.0 Console

File Edit Control Applications ?

File Browser

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Documents

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ex 6.sci

EXP 14.sci

Scilab 2026.1.0 Console

"Magnetic Field Intensity (A/m) Magnetic Flux Density (T)"

| Magnetic Field Intensity (A/m) | Magnetic Flux Density (T) |
|--------------------------------|---------------------------|
| 200                            | 0.251327                  |
| 500                            | 0.628319                  |
| 1000                           | 1.256637                  |
| 1500                           | 1.884956                  |
| 2000                           | 2.513274                  |

Variable Browser

| Name     | Value    | Type   | Visibility | Memory |
|----------|----------|--------|------------|--------|
| B_values | 1x5      | Double | local      | 248 B  |
| H        | 2e+03    | Double | local      | 216 B  |
| H_values | 1x5      | Double | local      | 216 B  |
| L        | 0.1      | Double | local      | 216 B  |
| W        | 0.05     | Double | local      | 216 B  |
| mu       | 5        | Double | local      | 216 B  |
| mu_0     | 0.00126  | Double | local      | 216 B  |
| mu_r     | 1.26e-06 | Double | local      | 216 B  |
| mu_r     | 1e+03    | Double | local      | 216 B  |

Command History

EXP 14.sci (C:\Users\user\OneDrive\Documents\EXP 14.sci) - SciNotes

File Edit Format Options Window Execute ?

EXP 14.sci X

1 clc;

2 clear;

3 clf();

4

5 // Define constants

6 L = 0.1; // Length in meters (10 cm)

7 W = 0.05; // Width in meters (5 cm)

8

9 H\_values = [200, 500, 1000, 1500, 2000]; // H in A/m

10

11 mu\_0 = 4 \* pi \* 10^(-7); // Permeability of free space (H/m)

12 mu\_r = 1000; // Relative permeability of soft iron

13

14 // Initialise array for magnetic flux density

15 B\_values = zeros(1, length(H\_values));

16

17 // Calculate B for each H value

18 for i = 1:length(H\_values)

19 H = H\_values(i);

20 mu = mu\_0 \* mu\_r; // Absolute permeability

21 B\_values(i) = mu \* H; // B = mu \* H

22 end

23

24 // Display results

25 disp('Magnetic Field Intensity (A/m) --- Magnetic Flux Density (T)');

26

27 for i = 1:length(H\_values)

28 fprintf('%10.0f --- %10.0f\n', H\_values(i), B\_values(i));

29 end

30

31 // Plot the results

32 plot(H\_values, B\_values, 'o');

33

34 xlabel('Magnetic Field Intensity H (A/m)');

35 ylabel('Magnetic Flux Density B (T)');

36 title('Magnetic Flux Density vs Magnetic Field Intensity for Soft Iron');

37

38 grid();

39

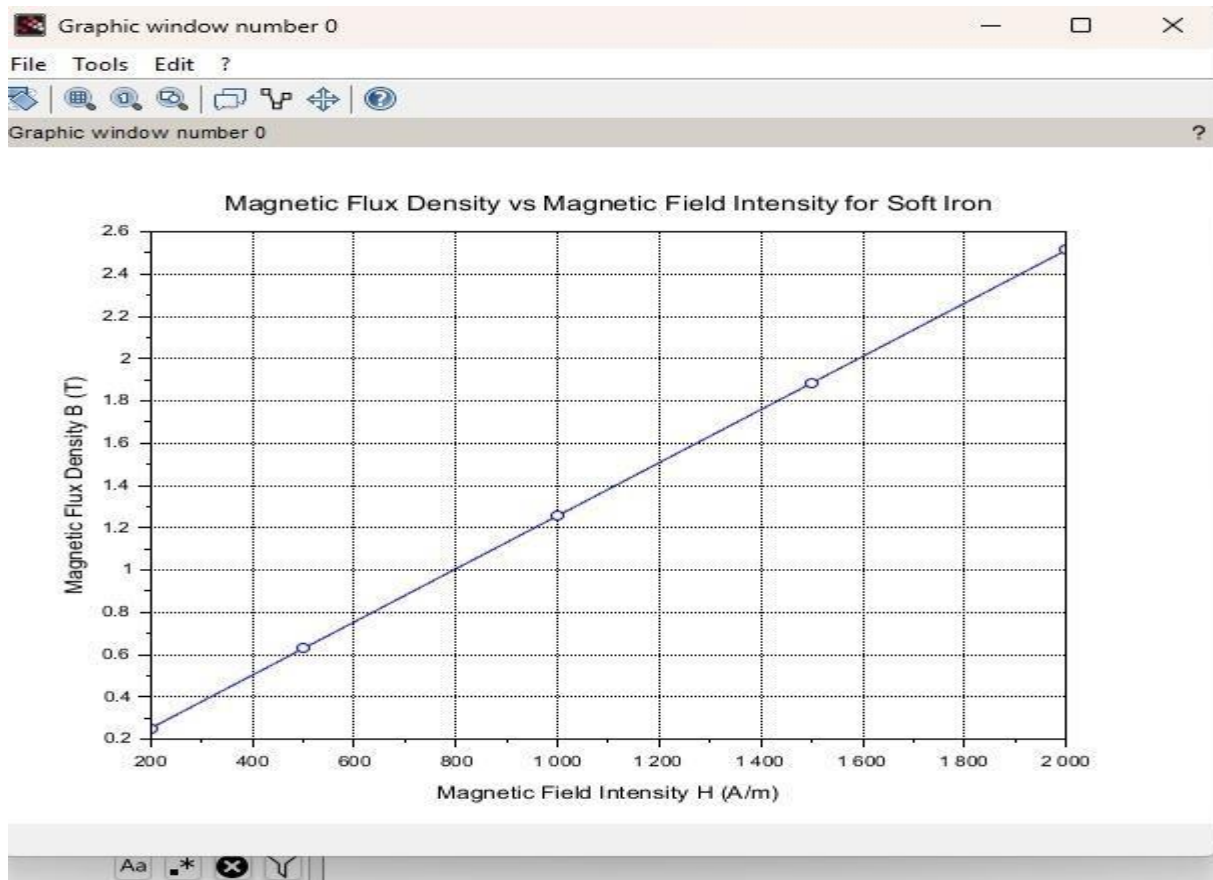
Snipping Tool

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11:33 AM 8/30/2026



**Result :** Thus the program was verified for program for Magnetic flux density in ferromagnetic materials using SCILAB was successfully.